Clinical Dataset Structure: A Universal Standard for Structuring Clinical Research Datasets

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CALIFORNIA MEDICAL INNOVATIONS INSTITUTE

FAIR DATA INNOVATIONS HUB

AI-READI

AI Ready and Exploratory Atlas for Diabetes Insights

Background

Background

AI-READI project

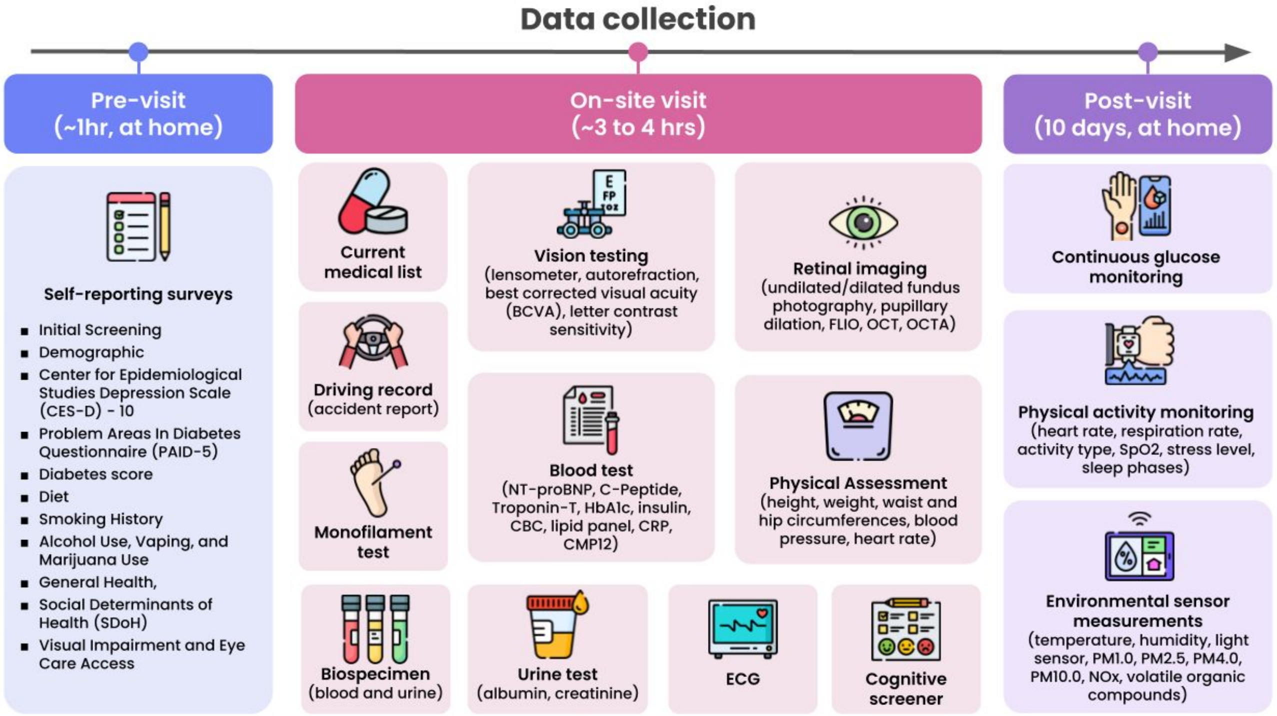
AI-READI is one of the Data Generation Project (DGP) of the NIH Bridge2AI Common Fund program (2022-Present)

Goal: Collect and share a multimodal dataset from 4,000 participants for studying Type 2 Diabetes

https://aireadi.org

Background

How to organize the multimodal AI-READI data?



FLIO = Fluorescence Lifetime Imaging; OCT = Optical Coherence Tomography; OCTA = Optical Coherence Tomography Angiography, ECG = Electrocardiogram; PM1.0, 2.5, 4.0, and 10.0 = Particulate matter less than 1, 2.5, 4, and 10 microns, respectively

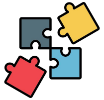
So many data files

→ How to organize the dataset such that it is interoperable and reusable?

Background
Challenge

There is currently no consensus on how to structure multimodal data into a consistently organized dataset in line with the FAIR principles

Datasets are hard to reuse
Difficult for researchers other than the original creators



Datasets are not interoperable
Undermines the development of AI models

Clinical Dataset Structure

Clinical Dataset Structure

About

The Clinical Dataset Structure (CDS) is a specification to consistently structure data and metadata for multimodal datasets

Timeline: 2023-Present

Vision: All clinical research datasets should “look” the same

https://cds-specification.readthedocs.io

Clinical Dataset Structure

Methods

01

Review & Adapt Existing Standards

Reviewed domain-specific dataset structures to leverage existing standards.

- Brain Imaging Data Structure
- SPARC Data Structure

02

Extend for Multimodal Datasets

Incorporated missing elements using existing schemas when possible.

- Study metadata → ClinicalTrials.gov schema
- Dataset metadata: DataCite schema

03

Implement & Document

Applied specification to a real-world clinical research dataset.

- AI-READI dataset implementation
- Detailed documentation to support reuse

Specification

Specification

Consistent folder structure

- cardiac_ecg
- clinical_data
- environment
- retinal_flio
- retinal_oct
- retinal_octa
- retinal_photography
- wearable_activity_monitor
- wearable_blood_glucose

One folder per datatype

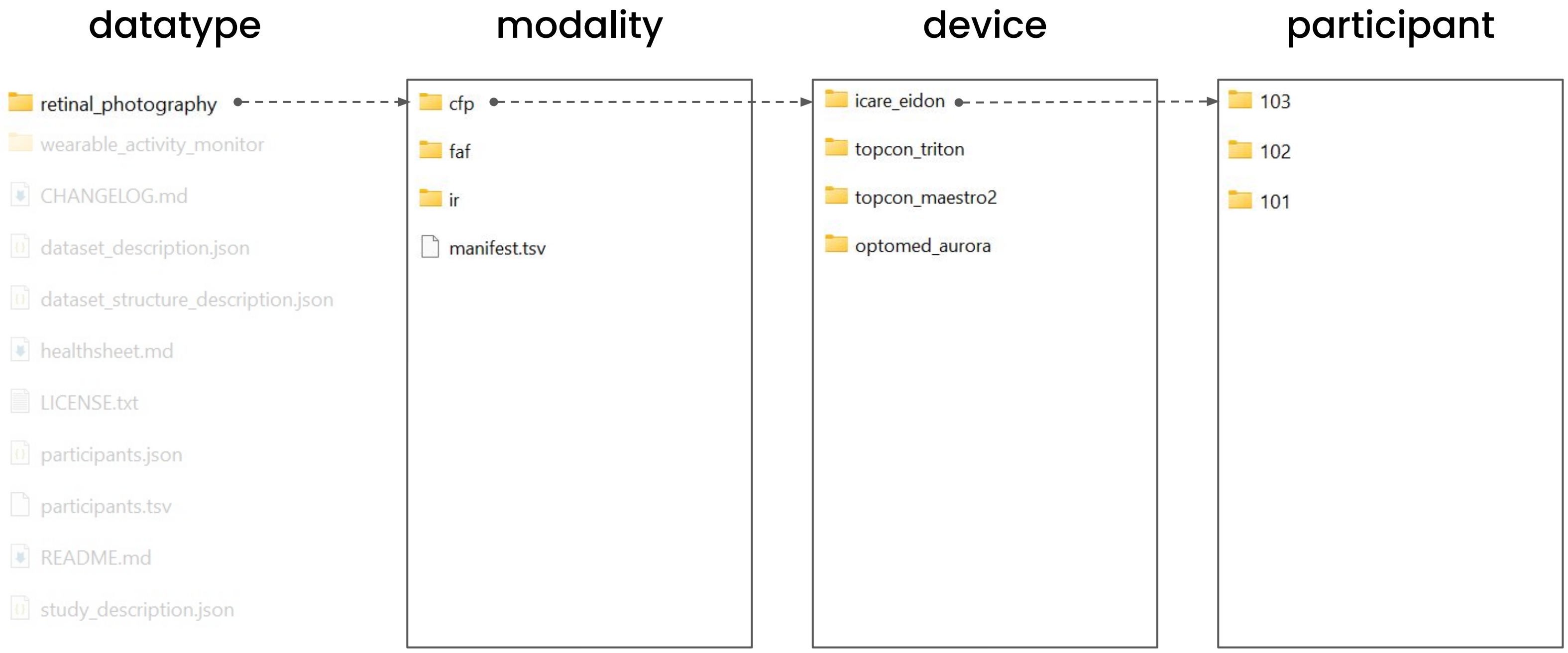
- CHANGELOG.md
- LICENSE.txt
- README.md
- dataset_description.json
- dataset_structure_description.json
- healthsheet.md
- participants.json
- participants.tsv
- study_description.json

Metadata files

Access the AI-READI dataset at fairhub.io

Specification

Consistent folder structure



Specification

Standard data format

Data category in AI-READI	Data format followed
Clinical data	Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM)
Retinal imaging data	Digital Imaging and Communications in Medicine (DICOM)
ECG data (standard 12-lead)	WaveForm DataBase (WFDB)
Wearables data (physical activity monitoring, continuous glucose monitoring data)	Open mHealth
Environmental sensor data	Earth Science Data Systems (ESDS) format

Specification

Extensive metadata

- retinal_photography
- wearable_activity_monitor
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- dataset_structure_description.json
- healthsheet.md
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- participants.tsv
- README.md
- study_description.json

Broad metadata: study background, dataset provenance, dataset structure, license terms, participants information

Provided in both human and machine-friendly formats

Specification

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- study_description.json


```
{
  "schema": "https://schema.aireadi.org/v0.1.1/dataset_structure_description.json",
  "directoryList": [
    {
      "directoryName": "cardiac_ecg",
      "directoryType": "dataType",
      "directoryDescription": "This directory contains electrocardiogram data collected by a 12 lead protocol
      "relatedIdentifier": [
        {
          "relatedIdentifierValue": "https://docs.aireadi.org",
          "relatedIdentifierType": "URL",
          "relationType": "IsDocumentedBy",
          "resourceTypeGeneral": "Other",
          "relatedIdentifierDescription": "This is the documentation of the AI-READI dataset that contains add
        }
      ],
      "relatedTerm": [
        {
          "relatedTermValue": "Electrocardiogram",
          "relatedTermIdentifier": [
            {
              "relatedTermClassificationCode": "C168186",
              "relatedTermScheme": "NCI Thesaurus (NCIT)",
              "relatedTermSchemeURI": "https://ncim.nci.nih.gov/",
              "relatedTermValueURI": "https://ncit.nci.nih.gov/ncitbrowser/pages/concept_details.jsf?dictionar
            }
```

Specification

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participant_id	study_group	age	study_visit_date	recommended_split	cardiac_ecg	clinical_data
101	pre_diabetes_lifestyle_controlled	51	8/22/2023	train	TRUE	TRUE
102	healthy	46	9/12/2023	train	TRUE	TRUE
103	oral_medication_and_or_non_insulin_injectable_medication_controlled	81	11/2/2023	test	TRUE	TRUE
104	oral_medication_and_or_non_insulin_injectable_medication_controlled	62	9/18/2023	val	TRUE	TRUE

Closing Remarks

Closing Remarks

Current status

The current specification is fully documented
https://cds-specification.readthedocs.io

We are continuing to make improvements
For instance: include a Croissant schema metadata file

We are developing tools to implement CDS
https://github.com/AI-READI/pyfairdatatools

The AI-READI dataset is being reused
Accessed by 1,000+ researchers

Closing Remarks

Next steps

Promote the CDS

We want to evaluate interoperability with other datasets

Collect feedback and improve

Planning a survey of the AI-READI dataset users

Closing Remarks
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Developers

Interns

Clinical Research Coordinators





Program and Project Managers

https://aireadi.org/team

Thank You!

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fairdataihub.org

Find these slides and all resources here

bit.ly/ISMB-CDS